Devixo Solutions

AI/ML Internship Program

Task 01

Python Programming, Data Analysis & Visualization

Project: Hotel Booking Demand Analysis

Student Name: Aman Hameed Ullah

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SECTION 1: Introduction

The hotel industry generates large amounts of booking data every day. Analyzing this data helps hotels understand guest behavior, predict cancellations, plan pricing and manage rooms more effectively. Booking cancellations are especially important, because an empty room that was already reserved means lost revenue.

This report documents Task 01 of the Devixo Solutions AI/ML Internship Program. It follows a complete data analysis workflow on a real-world hotel booking dataset using Python: loading and understanding the data, cleaning and preprocessing it, performing exploratory and statistical analysis, creating visualizations, drawing insights and building an interactive Streamlit dashboard.

SECTION 2: Objective

The objectives of this project are to:

- Load a real-world dataset using Python and Pandas and understand its structure.
- Identify and fix data quality problems such as missing values, duplicates, wrong data types and invalid records.
- Perform exploratory data analysis and statistical analysis.
- Create meaningful visualizations to reveal patterns and relationships.
- Generate useful insights and conclusions based on the actual data.
- Document the work professionally and organize it in a GitHub repository.

SECTION 3 TO 6: Dataset Details

Table 1: Dataset details

Item	Detail
Dataset Name	Hotel Booking Demand
Source	Kaggle: https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
Original Authors	Nuno Antonio, Ana de Almeida, Luis Nunes (published in Data in Brief, 2019)
Size (raw)	119,390 rows × 32 columns
File	hotel_bookings.csv
Period Covered	Arrivals from July 2015 to August 2017
Size (after cleaning)	86,611 rows × 35 columns

The dataset contains booking records of two hotels in Portugal: a City Hotel and a Resort Hotel. Each row represents one booking and describes when the booking was made, the arrival date, the length of stay, the number and type of guests, the meal plan, the booking channel, the price per

night and whether the booking was cancelled. The dataset includes both numerical and categorical variables, which makes it suitable for statistical analysis and visualization.

SECTION 7: Features / Columns Description

Table 2: Feature groups and column descriptions

Feature Group	Columns	Description
Booking Info	hotel, is_canceled, lead_time, reservation_status, reservation_status_date	Hotel type, cancellation flag (0/1), days between booking and arrival, last status and its date
Arrival Date	arrival_date_year, arrival_date_month, arrival_date_week_number, arrival_date_day_of_month	Year, month, week number and day of arrival
Length of Stay	stays_in_weekend_nights, stays_in_week_nights	Number of weekend and week nights booked
Guests	adults, children, babies, country, is_repeated_guest, customer_type	Guest counts, country of origin, repeat-guest flag (0/1), customer category
Booking Channel	market_segment, distribution_channel, agent, company (dropped)	How the booking was made; travel agent and company IDs (0 = none)
Rooms & Meals	meal, reserved_room_type, assigned_room_type	Meal plan, room type reserved and actually assigned
Guest History	previous_cancellations, previous_bookings_not_canceled, booking_changes	Earlier cancelled and completed bookings, number of changes to this booking
Pricing & Payment	adr, deposit_type, days_in_waiting_list	Average daily rate (price per night), deposit policy, waiting-list days
Requests	required_car_parking_spaces, total_of_special_requests	Parking spaces and special requests made

Note: column names shown are the original names. The renamed versions are listed in Section 10.4. The column company was dropped during cleaning because 94.3% of its values were missing.

SECTION 8: Methodology

The project followed a structured data analysis workflow:

Dataset Selection → Data Loading → Data Understanding → Data Cleaning → Exploratory Data Analysis → Statistical Analysis → Data Visualization → Insights → Conclusion

Tools and technologies used:

- Programming language: Python 3

- Environment: Google Colab (notebook) and VS Code (dashboard)
- Libraries: Pandas (data manipulation), NumPy (numerical operations), Matplotlib and Seaborn (visualization), Plotly and Streamlit (interactive dashboard)
- Storage and version control: Google Drive and GitHub

SECTION 9: Data Loading

The dataset was stored in Google Drive and loaded into a Pandas DataFrame using pd.read_csv(). The notebook was run in Google Colab after mounting Google Drive. Required libraries (Pandas, NumPy, Matplotlib and Seaborn) were imported first.

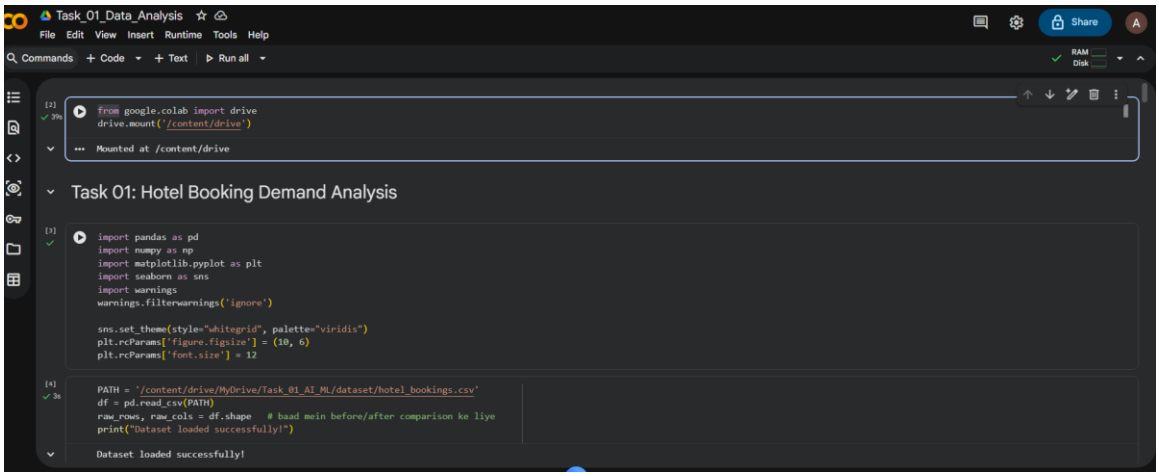


Figure 1: Import libraries and read_csv

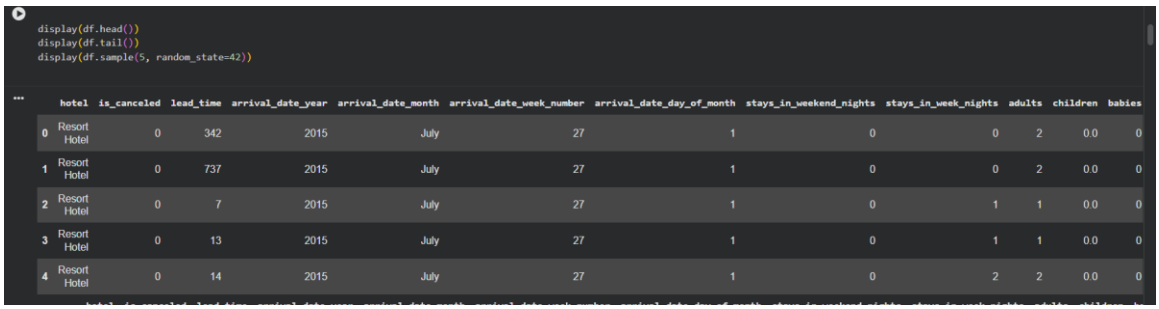


Figure 2: head() output



Figure 3: info() output

9.1 Understanding the Data

The raw dataset has 119,390 rows and 32 columns: 20 numerical columns and 12 categorical (text) columns. Only four columns contained missing values: company (112,593), agent (16,340), country (488) and children (4). The column reservation_status_date was stored as text and needed conversion to a date type.

SECTION 10: Data Cleaning

10.1 Missing Values

Four columns contained missing values. Their handling is summarized in Table 3 and the missing percentages are shown in Figure 4.

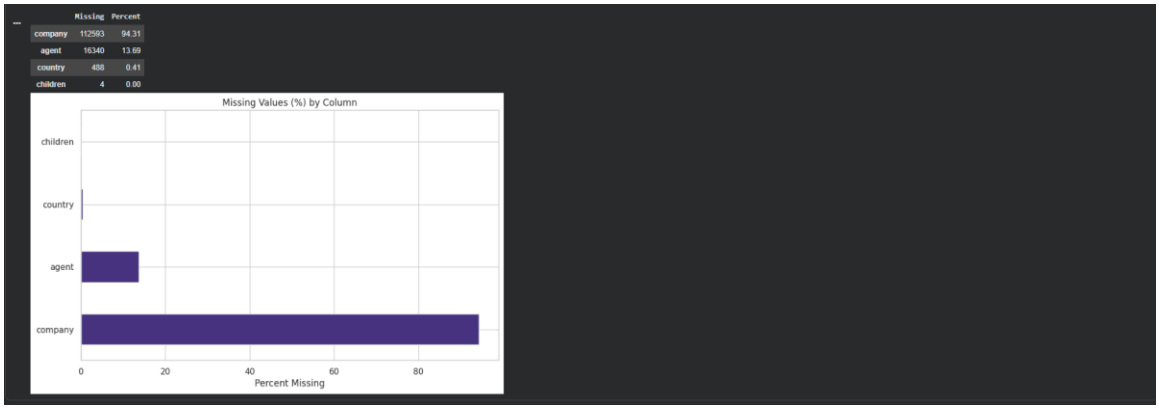


Figure 4: Missing values table and chart

Table 3: Handling of missing values

Column	Missing	Percent	Method	Reason
company	112,593	94.3%	Column dropped	Too sparse to be useful
agent	16,340	13.7%	Filled with 0	Missing means no agent was used
country	488	0.41%	Filled with mode	Categorical, very few missing
children	4	0.003%	Filled with 0	Most common value



Figure 5: Handling missing values

10.2 Duplicate Records

The dataset contained 32,020 duplicate records (about 26.8% of the data), meaning rows identical in every column. They were removed because duplicates give extra weight to the same booking and bias averages and cancellation rates. After removal, 87,370 rows remained and the duplicate count was 0.

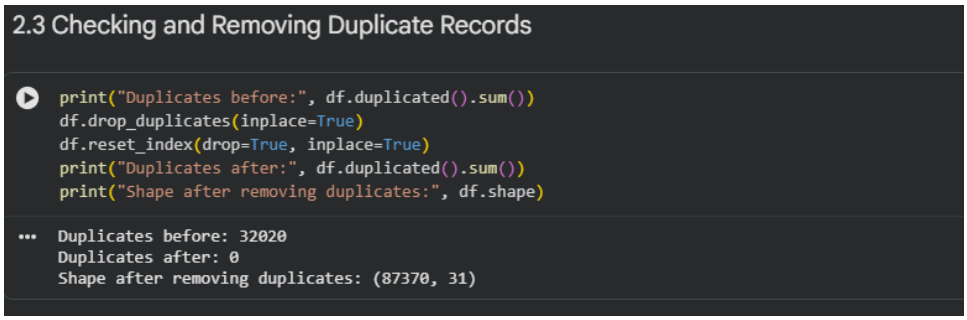


Figure 6: Duplicate checking and removal

10.3 Data Types

Some columns were stored in an unsuitable format. They were corrected as shown in Table 4. Two new columns were also created: arrival_date (built from year, month and day) for time-based analysis, and cancellation_status (Cancelled / Not Cancelled) for readable chart labels.

Table 4: Data type corrections

Column	Before	After
reservation_status_date	object (text)	datetime64
children	float64	int64
agent	float64	int64
arrival_date (new)	n/a	datetime64
cancellation_status (new)	n/a	object



Figure 7: Data types before and after correction

10.4 Renaming Columns

Several column names were long or unclear. They were renamed to be short, consistent and meaningful (Table 5).

Table 5: Renamed columns

Old Name	New Name
adr	avg_daily_rate
is_canceled	is_cancelled
arrival_date_year	arrival_year
arrival_date_month	arrival_month
arrival_date_week_number	arrival_week
arrival_date_day_of_month	arrival_day
stays_in_weekend_nights	weekend_nights
stays_in_week_nights	week_nights
reserved_room_type	reserved_room
assigned_room_type	assigned_room
required_car_parking_spaces	parking_spaces
total_of_special_requests	special_requests



Figure 8: List of columns after renaming

10.5 Invalid Values

Records that are logically impossible were identified and removed. Two helper columns, total_guests and total_nights, were created to perform these checks (Table 6).

Table 6: Invalid value checks

Check	Rows Found	Action
Negative average daily rate	1	Removed
Extreme rate (5,000 or more)	1	Removed
Zero guests	166	Removed
Zero nights stayed	651	Removed
Zero average daily rate	1,778	Kept (may be complimentary or special bookings)

In total, 759 rows were removed (some rows failed more than one check), leaving 86,611 records. These rows would have distorted averages and charts.

```
df['total_guests'] = df['adults'] + df['children'] + df['babies']
df['total_nights'] = df['weekend_nights'] + df['week_nights']

print("Negative rate:", (df.avg_daily_rate < 0).sum())
print("Zero rate:", (df.avg_daily_rate == 0).sum())
print("Extreme rate (>=5000):", (df.avg_daily_rate >= 5000).sum())
print("Zero guests:", (df.total_guests == 0).sum())
print("Zero nights:", (df.total_nights == 0).sum())

before = len(df)
df = df[(df.avg_daily_rate >= 0) & (df.avg_daily_rate < 5000)]
df = df[df.total_guests > 0]
df = df[df.total_nights > 0]
df.reset_index(drop=True, inplace=True)

print("Rows before:", before)
print("Rows after:", len(df))
print("Rows removed:", before - len(df))
```

```
Negative rate: 1
Zero rate: 1778
Extreme rate (>=5000): 1
Zero guests: 166
Zero nights: 651
Rows before: 87370
Rows after: 86611
Rows removed: 759
```

Figure 9: Invalid values check and removal

10.6 Final Verification

After cleaning, the dataset has no missing values, no duplicate records, suitable data types and no impossible values. Table 7 compares the dataset before and after cleaning.

Table 7: Dataset before and after cleaning

Metric	Before Cleaning	After Cleaning
Rows	119,390	86,611
Columns	32	35
Missing Values	4 columns affected	0
Duplicate Rows	32,020	0
Invalid Rows Removed	Not checked	759

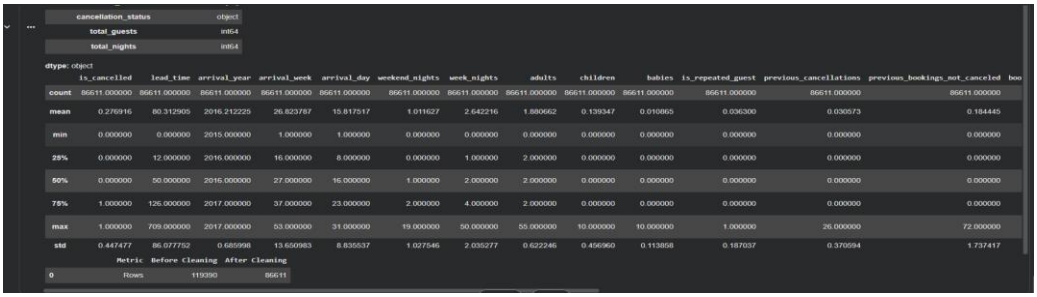


Figure 10: describe() output and before/after comparison

Overall, 32,779 rows (27.5% of the raw data) were removed, which ensures that all later analysis is based on reliable data.

SECTION 11: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed on the cleaned dataset to understand its contents, find the most frequent categories, detect unusual values and compare groups. All results below are based on the actual cleaned data of 86,611 bookings.

11.1 Dataset Overview

The cleaned dataset contains 86,611 bookings with arrival dates from 1 July 2015 to 31 August 2017. The City Hotel accounts for 53,040 bookings (61.24%) and the Resort Hotel for 33,571 bookings (38.76%). The overall cancellation rate is 27.69%, which means that roughly one in every four bookings is cancelled. The cancellation rate is higher at the City Hotel (30.21%) than at the Resort Hotel (23.72%).

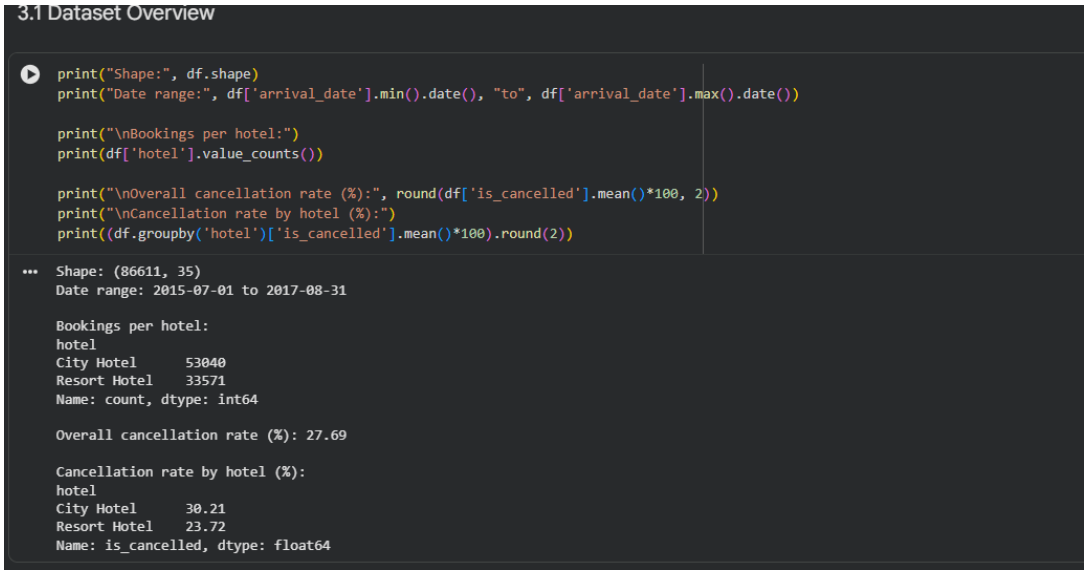


Figure 11: Dataset overview and cancellation rate by hotel

11.2 Frequency of Categories

Frequency counts show which categories dominate the data. Table 8 lists the most common value in the main categorical columns.

Table 8: Most frequent categories

Column	Most Frequent Value	Share of Bookings
meal	BB (Bed & Breakfast)	77.81%
market_segment	Online TA	59.21%
customer_type	Transient	82.38%
deposit_type	No Deposit	98.68%

Column	Most Frequent Value	Share of Bookings
country	PRT (Portugal)	31.50%

The most frequent value of a categorical column is its mode. The data shows that most bookings are made by transient customers (82.38%) through online travel agents (59.21%) without any deposit (98.68%), and that most guests choose bed and breakfast. Portugal is the largest source of guests (27,282 bookings), followed by Great Britain (10,399), France (8,813), Spain (7,228) and Germany (5,385).

11.3 Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. A value is treated as an outlier if it is below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR. The results are shown in Table 9.

Table 9: Outliers detected using the IQR method

Column	Lower Bound	Upper Bound	Outliers	Outlier %
lead_time	-159.00	297.00	2,353	2.72%
avg_daily_rate	-19.42	226.77	2,508	2.90%
total_nights	-2.50	9.50	2,990	3.45%
days_in_waiting_list	0.00	0.00	854	0.99%

All outliers are on the high side; the negative lower bounds simply mean that no value falls below them. The outliers were not removed. They are genuine bookings, such as reservations made far in advance (the maximum lead time is 709 days) or very high prices in peak season, and not data-entry errors. Removing them would hide real customer behavior. Instead, median-based measures are used where outliers could distort the mean.

11.4 Group-wise Analysis

Grouping the data by category reveals differences between customer groups. The main comparisons are summarized in Table 10.

Table 10: Key group-wise findings

Comparison	Result
Average daily rate by hotel	City Hotel: 111.66, Resort Hotel: 100.15
Highest average rate by market segment	Online TA: 118.91 (Direct: 118.08)
Highest cancellation rate by deposit type	Non Refund: 94.70%
Highest cancellation rate by market segment	Online TA: 35.55%
Average lead time: cancelled vs not cancelled	105.8 vs 70.5 days
Busiest month / most expensive month	August (11,194 bookings) / August (151.70)

The City Hotel has a higher average daily rate (111.66) than the Resort Hotel (100.15) and also a higher cancellation rate. Online TA bookings have the highest average rate and the highest cancellation rate (35.55%), while Corporate and Direct bookings cancel far less (12.23% and 14.88%). Cancelled bookings were made much earlier (105.8 days ahead) than completed bookings (70.5 days). Bookings and prices peak in August (11,194 bookings, average rate 151.70) and are lowest in January (4,636 bookings, average rate 70.89). The Undefined market segment shows 100% cancellation, but it contains only 2 bookings, so it is not meaningful. A surprising result is that Non Refund bookings have a 94.70% cancellation rate, even though guests pay in advance.

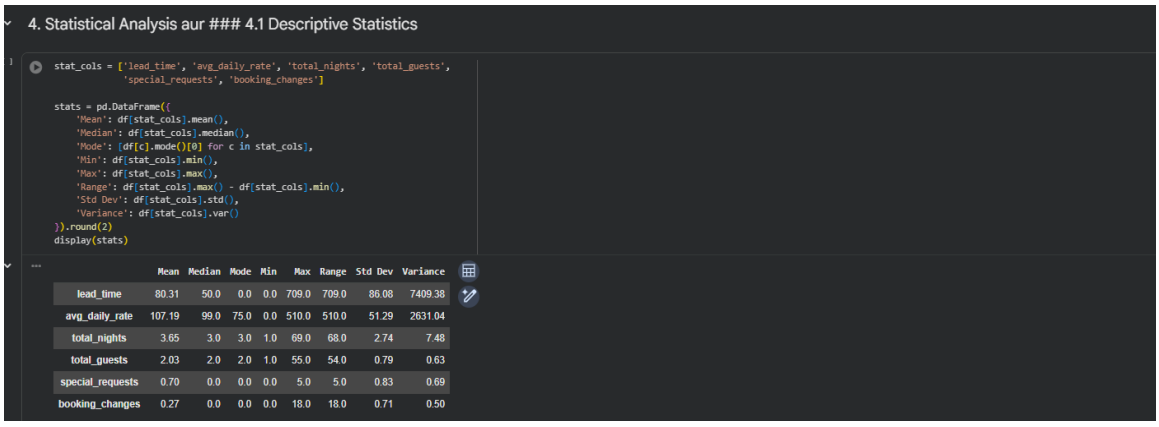


Figure 12: Group-wise analysis output

SECTION 12: Statistical Analysis

12.1 Descriptive Statistics

Descriptive statistics (mean, median, mode, minimum, maximum, range, standard deviation and variance) were calculated for the main numerical columns: lead_time, avg_daily_rate, total_nights, total_guests, special_requests and booking_changes.



	Mean	Median	Mode	Min	Max	Range	Std Dev	Variance
lead_time	80.31	50.0	0.0	0.0	709.0	709.0	86.08	7409.38
avg_daily_rate	107.19	99.0	75.0	0.0	510.0	510.0	51.29	2631.04
total_nights	3.65	3.0	3.0	1.0	69.0	68.0	2.74	7.48
total_guests	2.03	2.0	2.0	1.0	55.0	54.0	0.79	0.63
special_requests	0.70	0.0	0.0	0.0	5.0	5.0	0.83	0.69
booking_changes	0.27	0.0	0.0	0.0	18.0	18.0	0.71	0.50

Figure 13: Descriptive statistics of key numerical columns

For lead_time, the mean is 80.31 days while the median is 50 days, and values range from 0 to 709 days. The mean being higher than the median shows a right-skewed distribution. For avg_daily_rate, the mean is 107.19 and the median is 99.0, with a standard deviation of 51.29, so prices vary widely around a typical rate of about 100. A typical booking has 2 guests (mean 2.03, median 2) and stays 3 nights (mean 3.65, median 3). The mean of special_requests is 0.70 and of booking_changes is 0.27, and the median of both is 0, which shows that most guests make no special requests or changes.

12.2 Mean, Median and Mode

The mean is the average value and is calculated as:

$$\textbf{Mean} = \Sigma x / n$$

where Σx is the sum of all values and n is the number of observations. The mean of lead_time was calculated manually using this formula and compared with the Pandas result. Both gave 80.31 days, which verifies the calculation.

The median is the middle value of the sorted data, and the mode is the most frequent value. When extreme values exist, the median is more reliable than the mean. In this dataset the median lead time of 50 days describes a typical booking better than the mean of 80.31 days, because a few very early bookings pull the mean upwards. The mode of lead_time is 0 days, which means the most common booking is made on the same day as arrival.



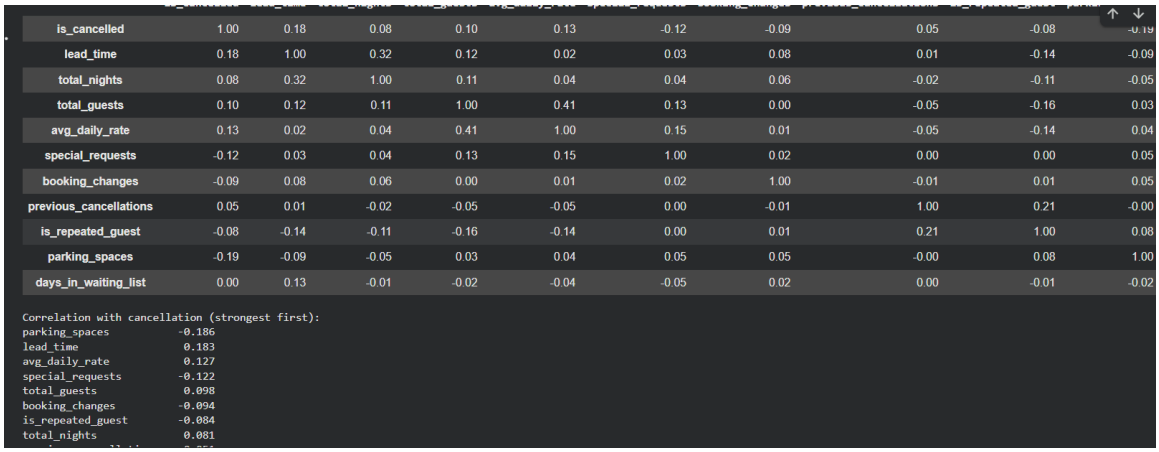
Figure 14: Manual mean calculation, median and mode

12.3 Standard Deviation and Variation

Standard deviation measures how far values spread around the mean; a larger value means greater variation. The lead_time standard deviation is 86.08 days, which is larger than its mean of 80.31 days, so guests differ greatly in how early they book. To compare variation between columns with different units, the coefficient of variation (standard deviation divided by mean) was used. booking_changes shows the highest relative variation (264.7%), followed by special_requests (118.9%) and lead_time (107.2%). total_guests is the most consistent column (38.9%), followed by avg_daily_rate (47.9%).

12.4 Correlation Analysis

A correlation matrix was calculated for the main numerical variables. The correlation coefficient lies between −1 and +1: +1 means a perfect positive relationship, 0 means no linear relationship and −1 means a perfect negative relationship.



is_cancelled	1.00	0.18	0.08	0.10	0.13	-0.12	-0.09	0.05	-0.08	-0.19
lead_time	0.18	1.00	0.32	0.12	0.02	0.03	0.08	0.01	-0.14	-0.09
total_nights	0.08	0.32	1.00	0.11	0.04	0.04	0.06	-0.02	-0.11	-0.05
total_guests	0.10	0.12	0.11	1.00	0.41	0.13	0.00	-0.05	-0.16	0.03
avg_daily_rate	0.13	0.02	0.04	0.41	1.00	0.15	0.01	-0.05	-0.14	0.04
special_requests	-0.12	0.03	0.04	0.13	0.15	1.00	0.02	0.00	0.00	0.05
booking_changes	-0.09	0.08	0.06	0.00	0.01	0.02	1.00	-0.01	0.01	0.05
previous_cancellations	0.05	0.01	-0.02	-0.05	-0.05	0.00	-0.01	1.00	0.21	-0.00
is_repeated_guest	-0.08	-0.14	-0.11	-0.16	-0.14	0.00	0.01	0.21	1.00	0.08
parking_spaces	-0.19	-0.09	-0.05	0.03	0.04	0.05	0.05	-0.00	0.08	1.00
days_in_waiting_list	0.00	0.13	-0.01	-0.02	-0.04	-0.05	0.02	0.00	-0.01	-0.02

```
Correlation with cancellation (strongest first):
parking_spaces          -0.186
lead_time                0.183
avg_daily_rate           0.127
special_requests        -0.122
total_guests             0.098
booking_changes         -0.094
is_repeated_guest       -0.084
total_nights             0.081
```

Figure 15: Correlation matrix of numerical variables

The most important findings are:

- parking_spaces has the strongest relationship with cancellation (r = −0.19, negative): guests who request a parking space are less likely to cancel. lead_time follows closely (r = 0.18, positive): the earlier a booking is made, the more likely it is to be cancelled.
- total_guests and avg_daily_rate show the strongest correlation in the data (r = 0.41, positive): bookings with more guests pay a higher daily rate. lead_time and total_nights are moderately related (r = 0.32): early bookings tend to be for longer stays.
- All correlations with cancellation are weak (the highest is 0.19 in absolute value), so no single variable can predict cancellation on its own.

Correlation shows how variables move together but does not prove that one causes the other. The column reservation_status was excluded because it is another form of the cancellation result and would give a misleading correlation.

SECTION 13: Data Visualizations

Thirteen visualizations were created with Matplotlib and Seaborn: the ten required chart types and three additional charts. Each chart has a title, axis labels and a short interpretation, and is saved at 300 dpi in the images folder of the GitHub repository.

13.1 Histogram: Distribution of Lead Time

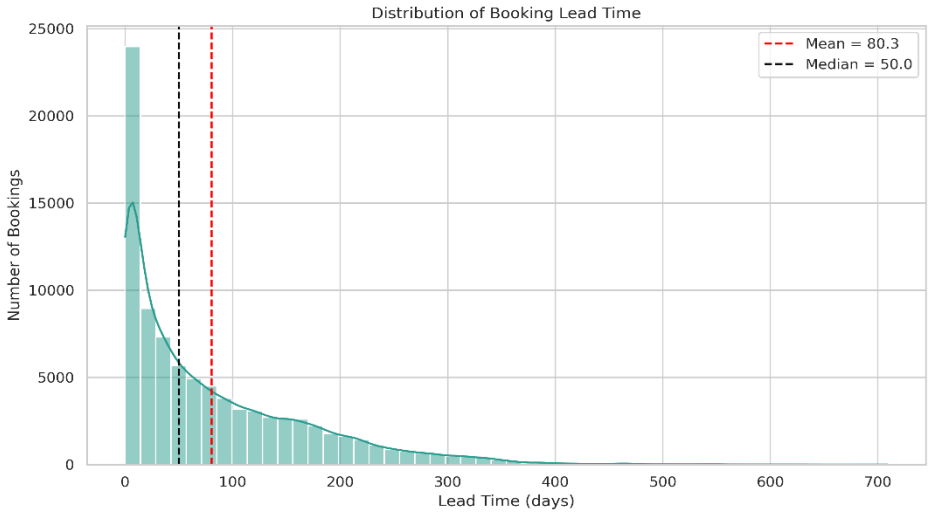


Figure 16: Distribution of lead time (histogram)

The histogram shows how many days before arrival guests book. The distribution is right-skewed: most bookings are made at short notice, while a few are made more than a year ahead (maximum 709 days). The mean (80.31 days) is above the median (50 days) because of these early bookings, so hotels cannot assume that guests plan far in advance.

13.2 Bar Chart: Average Daily Rate by Market Segment

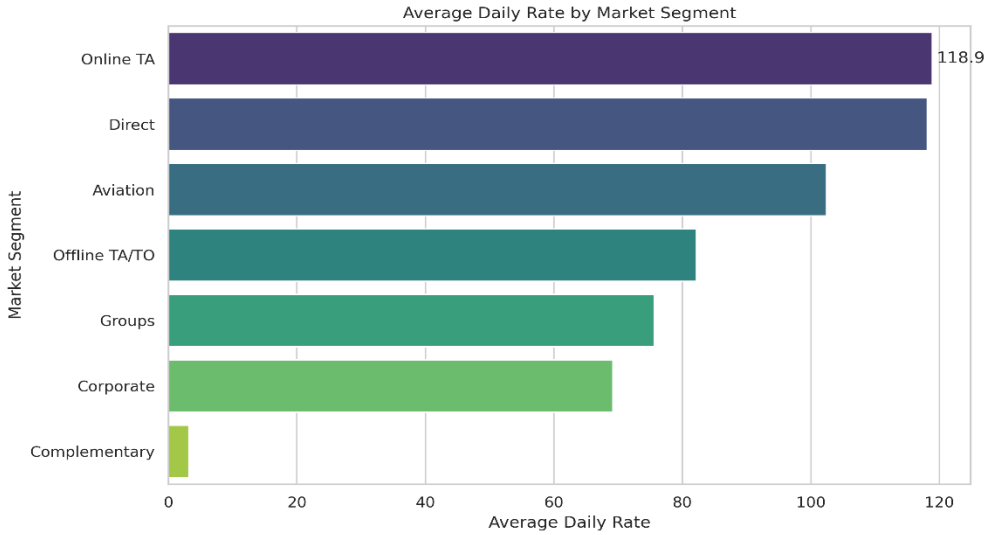


Figure 17: Average daily rate by market segment (bar chart)

Online TA (118.91) and Direct (118.08) have the highest average rates, while Corporate (69.17) and Groups (75.60) have the lowest. Individual guests pay near-full prices, whereas corporate and group bookings receive negotiated discounts. Hotels can use this to decide which channels bring the most revenue per night.

13.3 Pie Chart: Share of Bookings by Market Segment

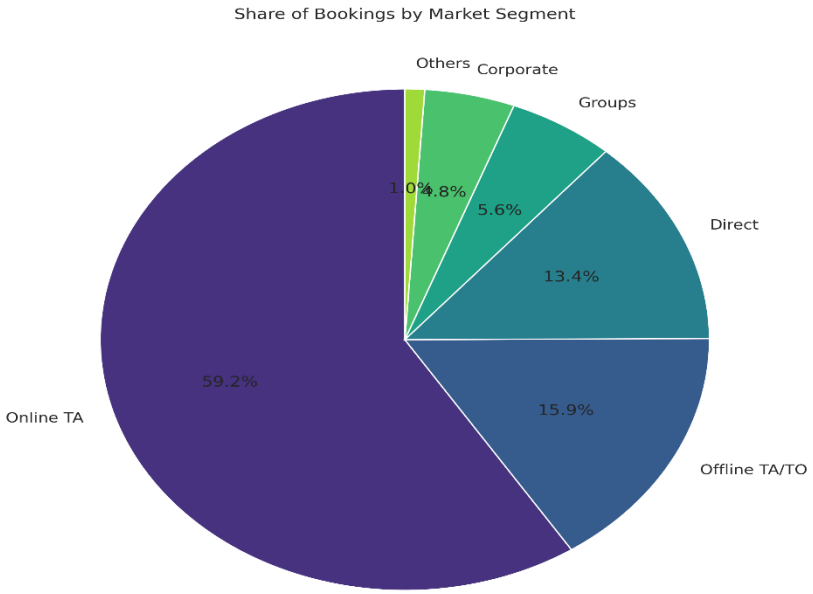


Figure 18: Share of bookings by market segment (pie chart)

Online TA accounts for about 59% of all bookings, followed by Offline TA/TO (about 16%) and Direct (about 13%). The hotels depend heavily on online travel agents, which is also the channel with the highest cancellation rate.

13.4 Scatter Plot: Total Guests vs Average Daily Rate

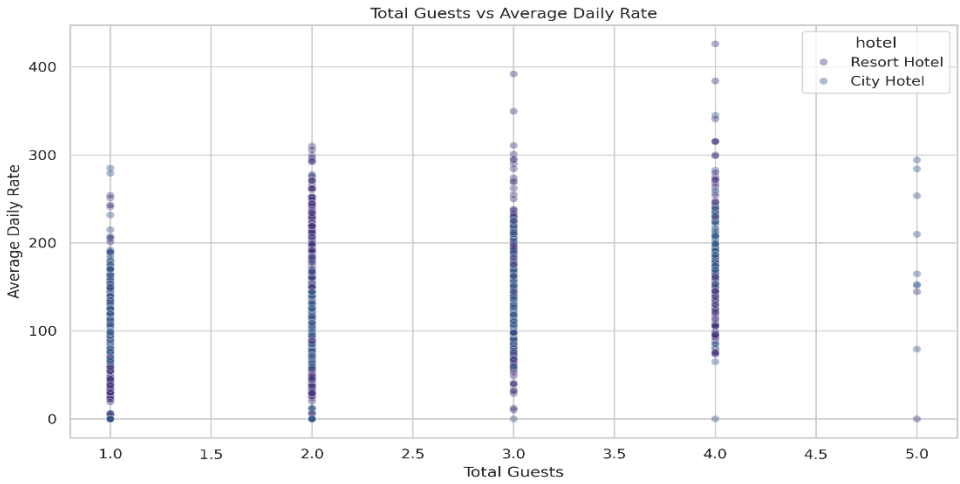


Figure 19: Total guests vs average daily rate (scatter plot)

Bookings with more guests tend to have a higher daily rate, which agrees with the correlation of 0.41, the strongest in the dataset. The points are widely spread, which shows that other factors such as season, hotel and room type also affect the price.

13.5 Line Chart: Average Daily Rate by Month

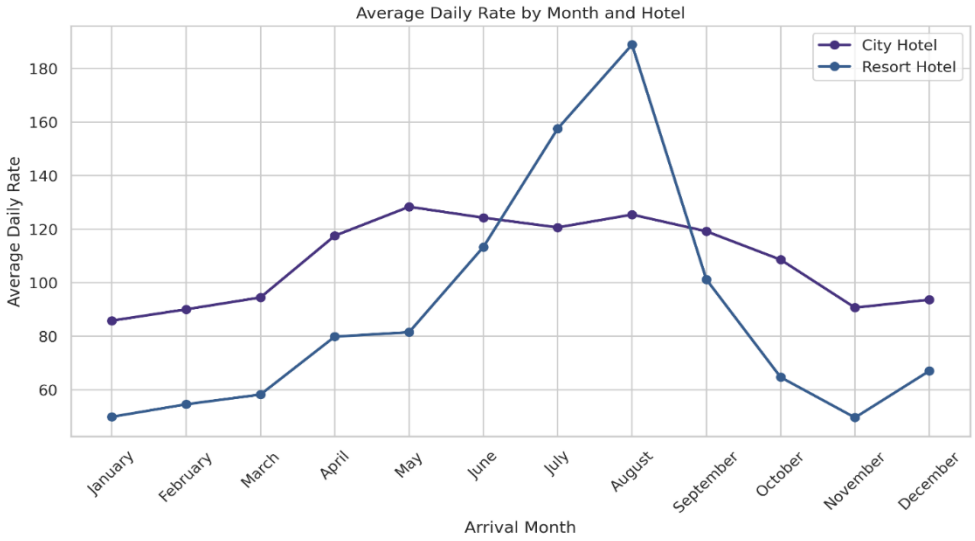


Figure 20: Average daily rate by month (line chart)

Prices rise from about 71 in January to a peak of about 152 in August and then fall again towards winter, which shows a clear seasonal pattern. Summer is the high season and winter the low season. Hotels can use this for pricing, staffing and promotions.

13.6 Box Plot: Average Daily Rate by Hotel

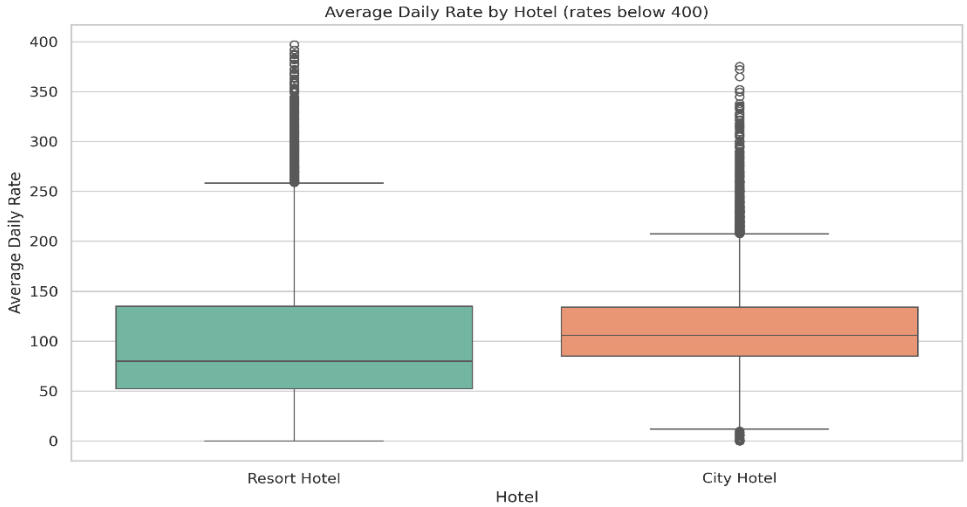


Figure 21: Average daily rate by hotel (box plot)

The City Hotel has a higher average rate (111.66) than the Resort Hotel (100.15). The points above the whiskers are outliers, which are high-priced bookings. They were kept because they are genuine bookings and not errors.

13.7 Correlation Heatmap

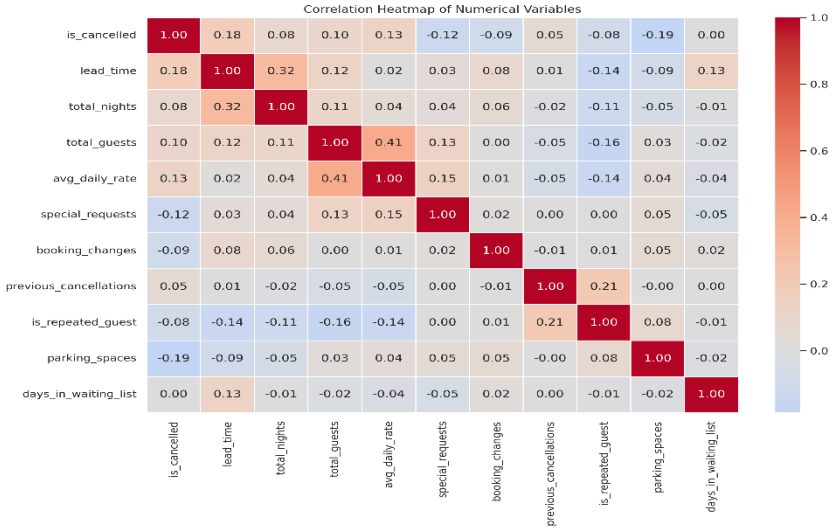


Figure 22: Correlation heatmap

The strongest correlation is between total guests and average daily rate (0.41), followed by lead time and total nights (0.32). All correlations with cancellation are weak (parking_spaces −0.19, lead_time 0.18), so no single variable can predict cancellation on its own.

13.8 Count Plot: Market Segment and Cancellation Status

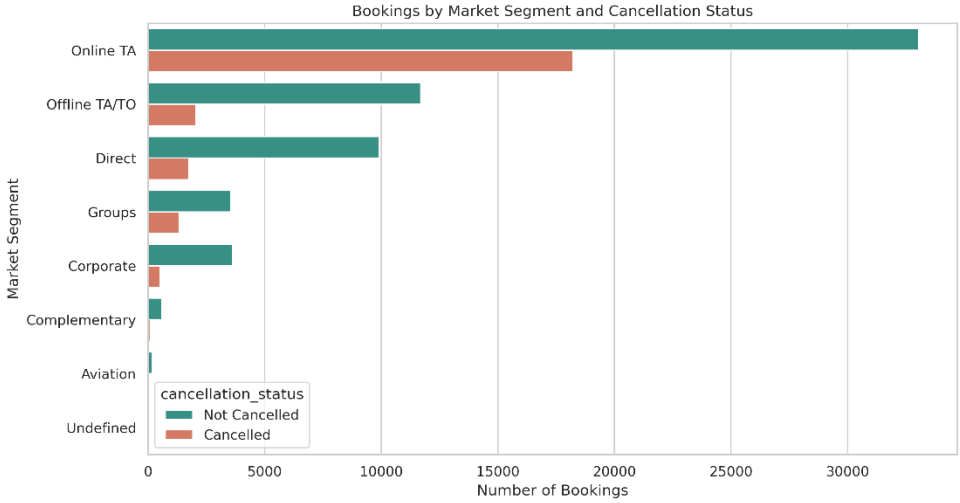


Figure 23: Bookings by market segment and cancellation status (count plot)

Online TA has by far the most bookings and also the highest cancellation rate (35.55%), while Corporate (12.23%) and Direct (14.88%) cancel much less. Online bookings are easy to make and to cancel, so they carry more risk for the hotels.

13.9 Pair Plot of Key Variables

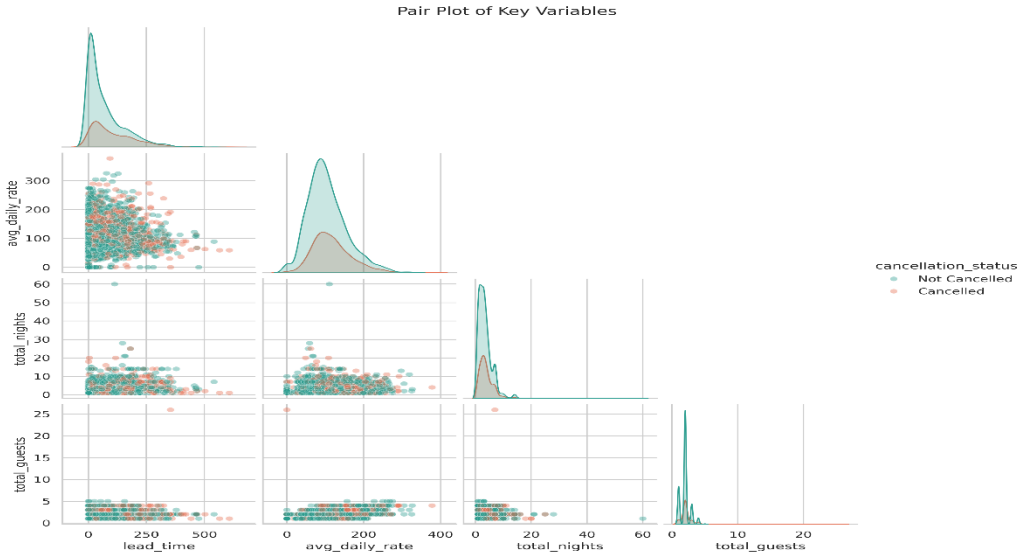


Figure 24: Pair plot of key variables

The pair plot shows the relationships between lead time, daily rate, total nights and total guests, coloured by cancellation status. Most pairs show weak relationships, which matches the heatmap. Cancelled and non-cancelled bookings overlap widely, so these variables alone cannot separate them.

13.10 Violin Plot: Lead Time by Cancellation Status

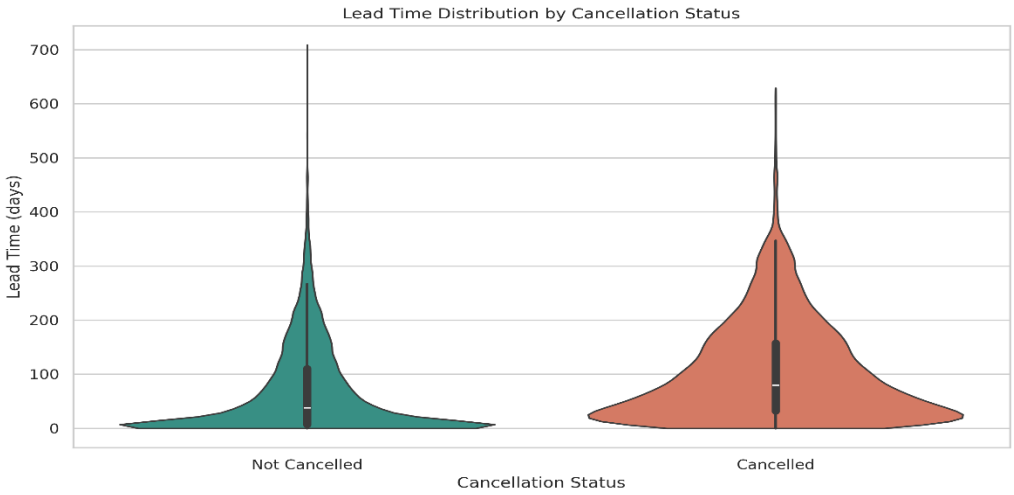


Figure 25: Lead time by cancellation status (violin plot)

Cancelled bookings were made much earlier (average 105.8 days) than non-cancelled bookings (average 70.5 days). Bookings made far in advance have more time to be cancelled because plans change, so hotels can watch long lead-time bookings more closely.

13.11 Count Plot: Bookings by Arrival Month

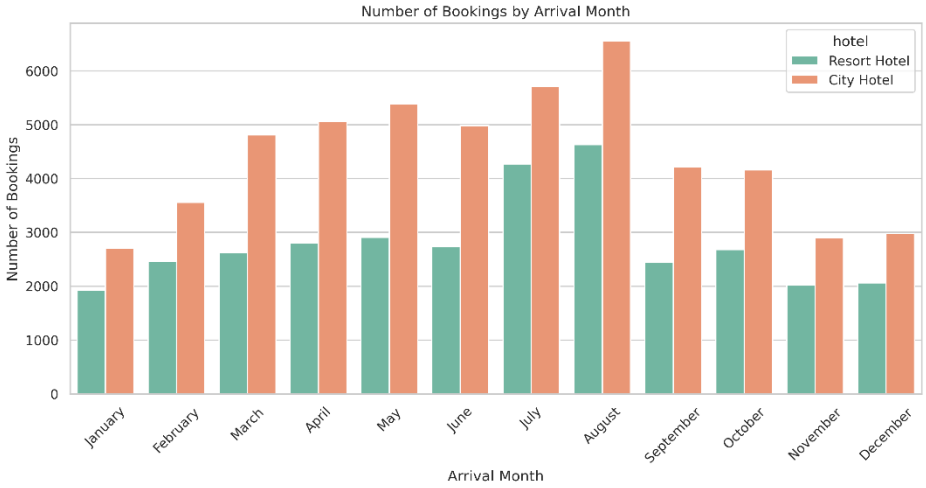


Figure 26: Bookings by arrival month (count plot)

August is the busiest month (11,194 bookings) and January the quietest (4,636). Bookings rise through spring, peak in summer and fall in autumn and winter, which matches the price pattern in Figure 20.

13.12 Bar Chart: Cancellation Rate by Deposit Type

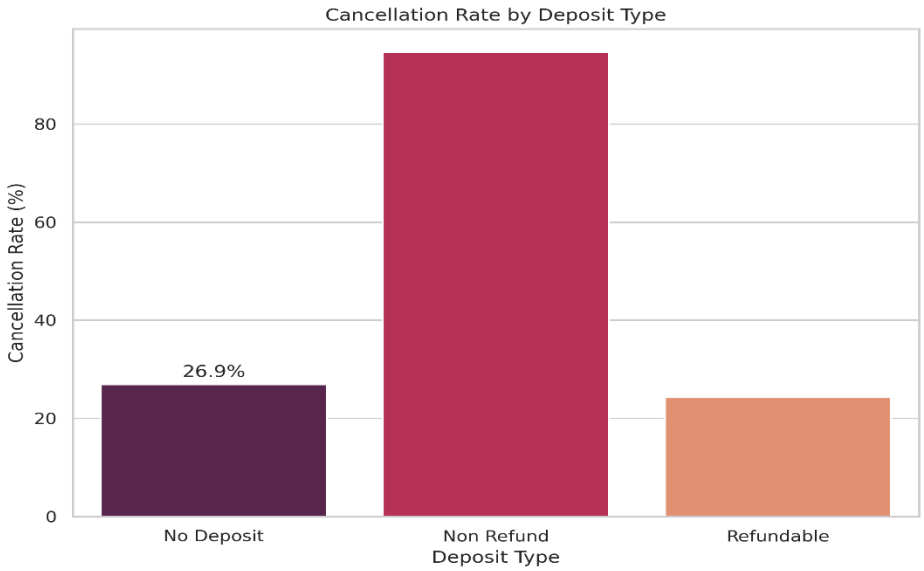


Figure 27: Cancellation rate by deposit type (bar chart)

Non Refund bookings have a cancellation rate of 94.70%, compared with 26.88% for No Deposit and 24.30% for Refundable. This is unexpected, because guests who pay in advance would be expected to cancel less, so the result should be treated with caution.

13.13 Bar Chart: Top 10 Guest Countries

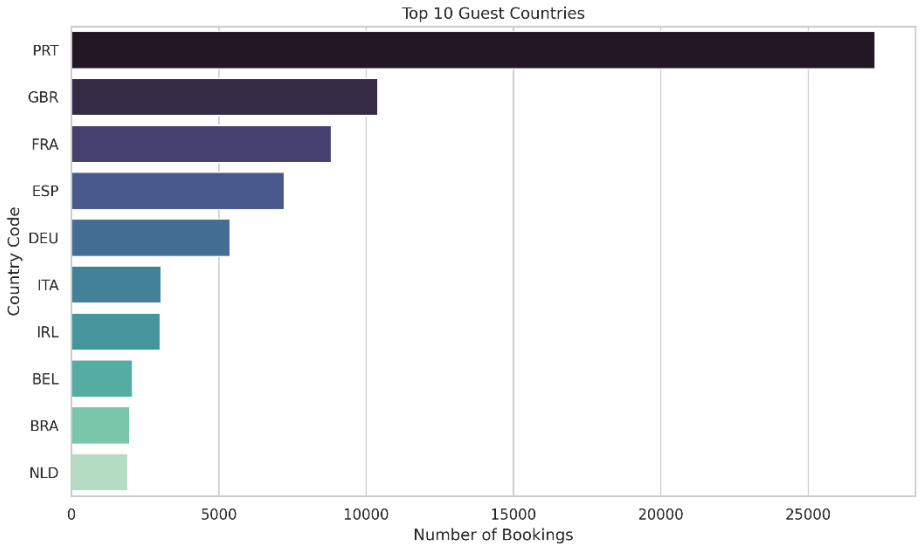


Figure 28: Top 10 guest countries (bar chart)

Portugal (PRT) is the largest source of guests with 27,282 bookings, followed by Great Britain (10,399), France (8,813), Spain (7,228) and Germany (5,385). Most guests come from Portugal and nearby European countries.

SECTION 14: Code Screenshots

The code for each step is shown together with its output in the figures listed in Table 11. The code used to create and save the visualizations is shown in Figure 29.

Table 11: Where to find the code screenshots

Step	Figure
Importing libraries and loading the dataset	Figure 1
Handling missing values	Figure 5
Removing duplicate records	Figure 6
Correcting data types	Figure 7
Renaming columns	Figure 8
Removing invalid values	Figure 9
Descriptive statistics	Figure 13
Manual mean, median and mode	Figure 14
Creating and saving a visualization	Figure 29

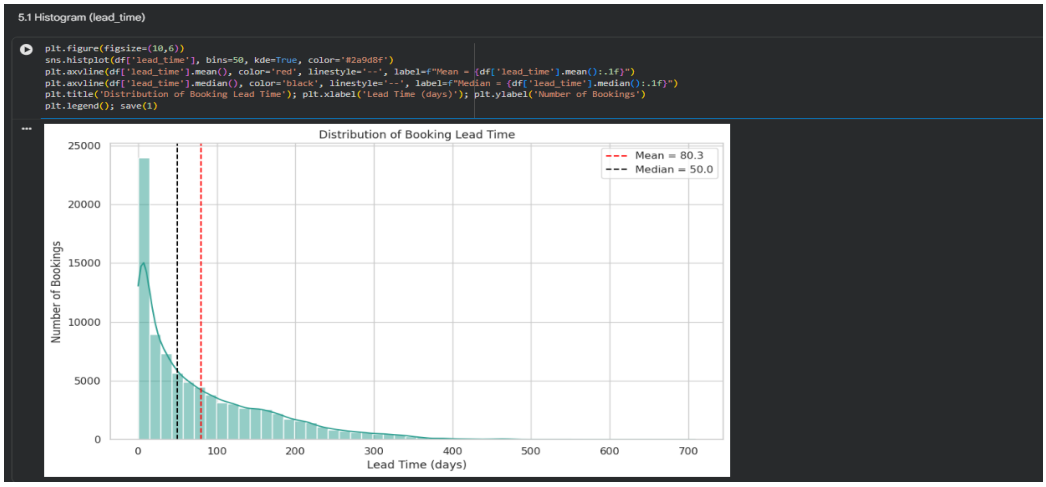


Figure 29: Code used to create and save a visualization

SECTION 15: Output Screenshots

The outputs of the main steps are shown in the figures listed in Table 12.

Table 12: Where to find the output screenshots

Output	Figure
Dataset records (head)	Figure 2
Dataset information (info)	Figure 3
Missing values table and chart	Figure 4
Data types before and after	Figure 7
Cleaned dataset verification and comparison	Figure 10
EDA results (overview and group-wise analysis)	Figures 11 and 12
Statistical analysis (statistics, mean, median, mode)	Figures 13 and 14
Correlation matrix	Figure 15
Streamlit dashboard	Figures 30 to 33

SECTION 16: Graph Screenshots

All 13 graphs are shown in Section 13 (Figures 16 to 28) and are also saved in the images folder of the GitHub repository, as listed in Table 13.

Table 13: Graph files in the repository

Figure	File Name	Chart
Figure 16	visualization_01.png	Histogram
Figure 17	visualization_02.png	Bar chart
Figure 18	visualization_03.png	Pie chart

Figure	File Name	Chart
Figure 19	visualization_04.png	Scatter plot
Figure 20	visualization_05.png	Line chart
Figure 21	visualization_06.png	Box plot
Figure 22	visualization_07.png	Heatmap
Figure 23	visualization_08.png	Count plot
Figure 24	visualization_09.png	Pair plot
Figure 25	visualization_10.png	Violin plot
Figure 26	visualization_11.png	Count plot
Figure 27	visualization_12.png	Bar chart
Figure 28	visualization_13.png	Bar chart

BONUS TASK: Streamlit Dashboard

An interactive dashboard was built with Streamlit and Plotly (app/app.py) to explore the cleaned dataset. It reads cleaned_hotel_bookings.csv and updates all numbers and charts when the filters are changed. The dashboard includes:

- Six KPI cards: total bookings, total columns, cancellation rate, average daily rate, average lead time and average stay.
- Sidebar filters: hotel, arrival year, market segment, average daily rate range and cancellation status.
- Four tabs: Overview (sample records and summary statistics), Charts, Correlation heatmap and Key Insights.
- Seven interactive charts: bookings by month, rate by month, cancellation rate by segment, lead time distribution, guests vs rate, top countries and market segment share.
- A download button for the filtered data.

To verify the dashboard, the hotel filter was set to Resort Hotel. It then shows 33,571 bookings and a cancellation rate of 23.72%, which matches the notebook results.

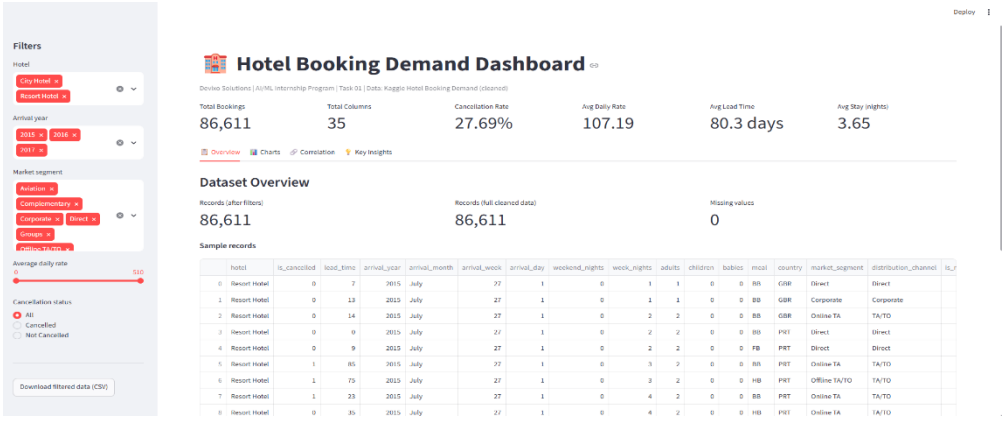


Figure 30: Dashboard overview with KPIs and filters

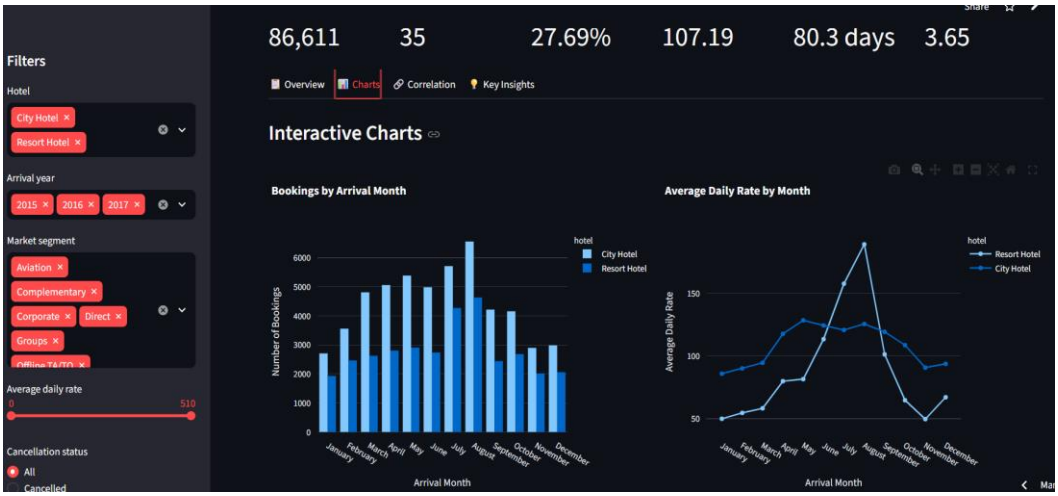


Figure 31: Interactive charts tab

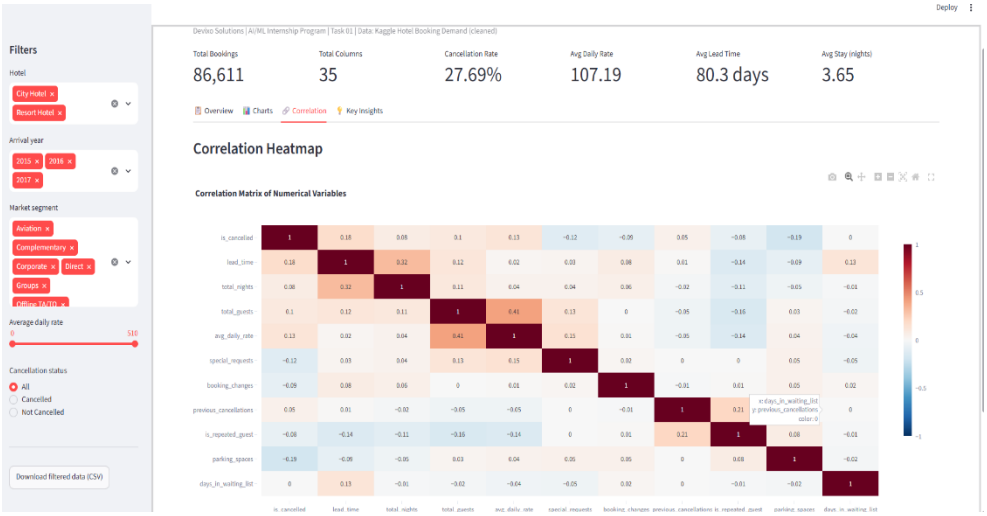


Figure 32: Correlation heatmap tab

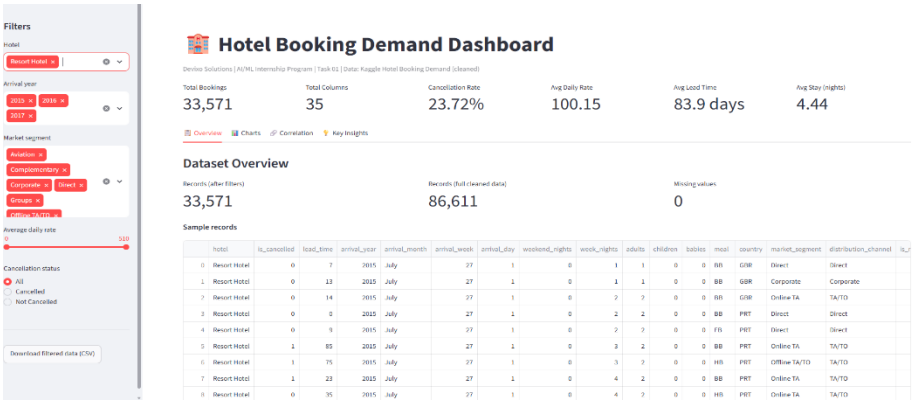


Figure 33: Dashboard with the Resort Hotel filter applied

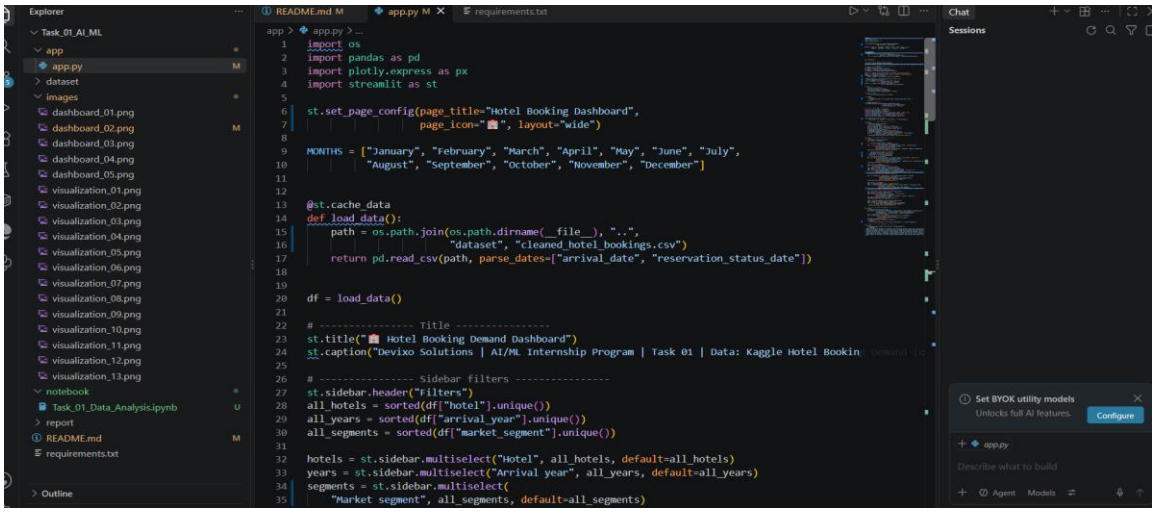


Figure 34: Streamlit code (app.py)

Live dashboard: https://khan-hotelbookinganalysis-m4hcvw26fs4byavnhowerk01.streamlit.app/

SECTION 17: Major Findings

- 27.69% of all bookings are cancelled; the City Hotel (30.21%) cancels more than the Resort Hotel (23.72%).
- Online TA brings 59.21% of bookings but has the highest cancellation rate (35.55%) among the main segments; Corporate (12.23%) and Direct (14.88%) are the most reliable.
- Cancelled bookings were made earlier (105.8 days ahead) than completed ones (70.5 days).
- Demand and prices peak in August (11,194 bookings, average rate 151.70) and are lowest in January (4,636 bookings, average rate 70.89).
- The strongest correlation is between total guests and average daily rate (0.41); all correlations with cancellation are weak (below 0.20).
- The median lead time (50 days) describes a typical booking better than the mean (80.31 days) because the data is right-skewed.

SECTION 18: Insights

The following 15 insights are based on the actual analysis of the cleaned dataset.

1. About one in four bookings is cancelled. The overall cancellation rate is 27.69%. Every cancelled booking is a room that may stay empty, so cancellations directly reduce revenue.

2. The City Hotel has more cancellations than the Resort Hotel. The rate is 30.21% for the City Hotel and 23.72% for the Resort Hotel. City guests, often on business or short trips, change plans more easily than holiday guests.

3. Online travel agents bring the most bookings and the most cancellations. Online TA makes up 59.21% of bookings and has a cancellation rate of 35.55%, so the hotels depend on the least reliable channel.

4. Corporate and Direct bookings are the most reliable. Their cancellation rates are 12.23% and 14.88%, less than half of the Online TA rate. Encouraging direct bookings can reduce losses.

5. Bookings made earlier are more likely to be cancelled. Cancelled bookings were made 105.8 days ahead on average, compared with 70.5 days for completed ones. Lead time also correlates with cancellation ($r = 0.18$).

6. Lead time is highly skewed. The mean is 80.31 days, the median is 50 days and the mode is 0 (same-day bookings). The maximum is 709 days, so the median describes a typical booking better.

7. Demand and price show strong seasonality. August has the most bookings (11,194) and the highest rate (151.70), while January has the fewest (4,636) and the lowest rate (70.89). The August rate is more than twice the January rate.

8. The City Hotel charges more than the Resort Hotel. The average daily rate is 111.66 against 100.15, about 11.5% higher, and the City Hotel also has more bookings (53,040 vs 33,571).

9. The number of guests is the strongest driver of price. Total guests and daily rate have the strongest correlation ($r = 0.41$). Lead time and total nights come next ($r = 0.32$): early bookings tend to be longer stays.

10. Guests who ask for parking or special services cancel less. parking_spaces ($r = -0.19$) and special_requests ($r = -0.12$) are negatively related to cancellation, which suggests that guests who plan their stay in detail are more committed.

11. Online and Direct bookings pay the most; Corporate and Groups pay the least. Online TA averages 118.91 and Direct 118.08, while Corporate averages 69.17, about 42% lower than Online TA.

12. Almost all bookings have no deposit. No Deposit makes up 98.68% of bookings. Unexpectedly, the small Non Refund group has a 94.70% cancellation rate against 26.88% for No Deposit, which is worth investigating.

13. Most guests are transient European travelers. Transient customers make up 82.38% of bookings and Portugal supplies 31.5%, followed by Great Britain, France, Spain and Germany. A typical booking has 2 guests and 3 nights.

14. Variation differs widely between variables. booking_changes has the highest coefficient of variation (264.7%) and total_guests the lowest (38.9%): changes are rare but uneven, while group size is quite uniform.

15. Outliers are genuine and limited. Only 2.72% of lead times, 2.90% of daily rates and 3.45% of stay lengths are outliers. They are real unusual bookings and were kept.

Overall, no single variable strongly predicts cancellation (all correlations are below 0.20). Cancellation depends on a combination of factors such as channel, hotel, lead time and guest behavior, which is why machine learning models will be useful in later tasks.

SECTION 19: Conclusion

In this task, the Hotel Booking Demand dataset from Kaggle was analyzed. It contains booking records of a City Hotel and a Resort Hotel in Portugal, which makes it suitable for studying guest behavior, pricing and cancellations. The raw data had 119,390 rows and 32 columns. During cleaning, the column company (94% missing) was dropped, missing values in agent, country and children were filled, 32,020 duplicate records were removed, data types were corrected, columns were renamed and 759 invalid rows (negative or extreme rates, zero guests, zero nights) were removed. The final dataset has 86,611 rows and 35 columns.

Exploratory and statistical analysis included descriptive statistics, mean, median, mode, standard deviation, variance, outlier detection with the IQR method, group-wise comparisons and correlation analysis, supported by 13 visualizations and an interactive Streamlit dashboard. The most important findings are that 27.69% of bookings are cancelled, that cancellations are highest at the City Hotel and for Online TA bookings, that earlier bookings cancel more often, and that prices and demand peak strongly in summer. All correlations with cancellation are weak, which shows that predicting cancellations needs more than one variable. Through this project I learned how to prepare a real-world dataset, justify cleaning decisions, interpret statistics and charts, and turn numbers into insights that can support business decisions such as pricing, channel strategy and overbooking policy.

SECTION 20: GitHub Repository Link

Table 14: Project links

Item	Link
GitHub Repository	https://github.com/Amankhan987/Task_01_AI-ML_Hotel_Booking_Analysis
Live Streamlit Dashboard	https://khan-hotelbookinganalysis-m4hcvw26fs4byavnhowerk01.streamlit.app/
Dataset Source (Kaggle)	https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand

The repository contains the Jupyter notebook, the raw and cleaned datasets, all visualization images, this PDF report, the Streamlit application, requirements.txt and README.md.